

Needle-Free

INJECTOR

Technology And Competitor
Landscape Report

SAMPLE

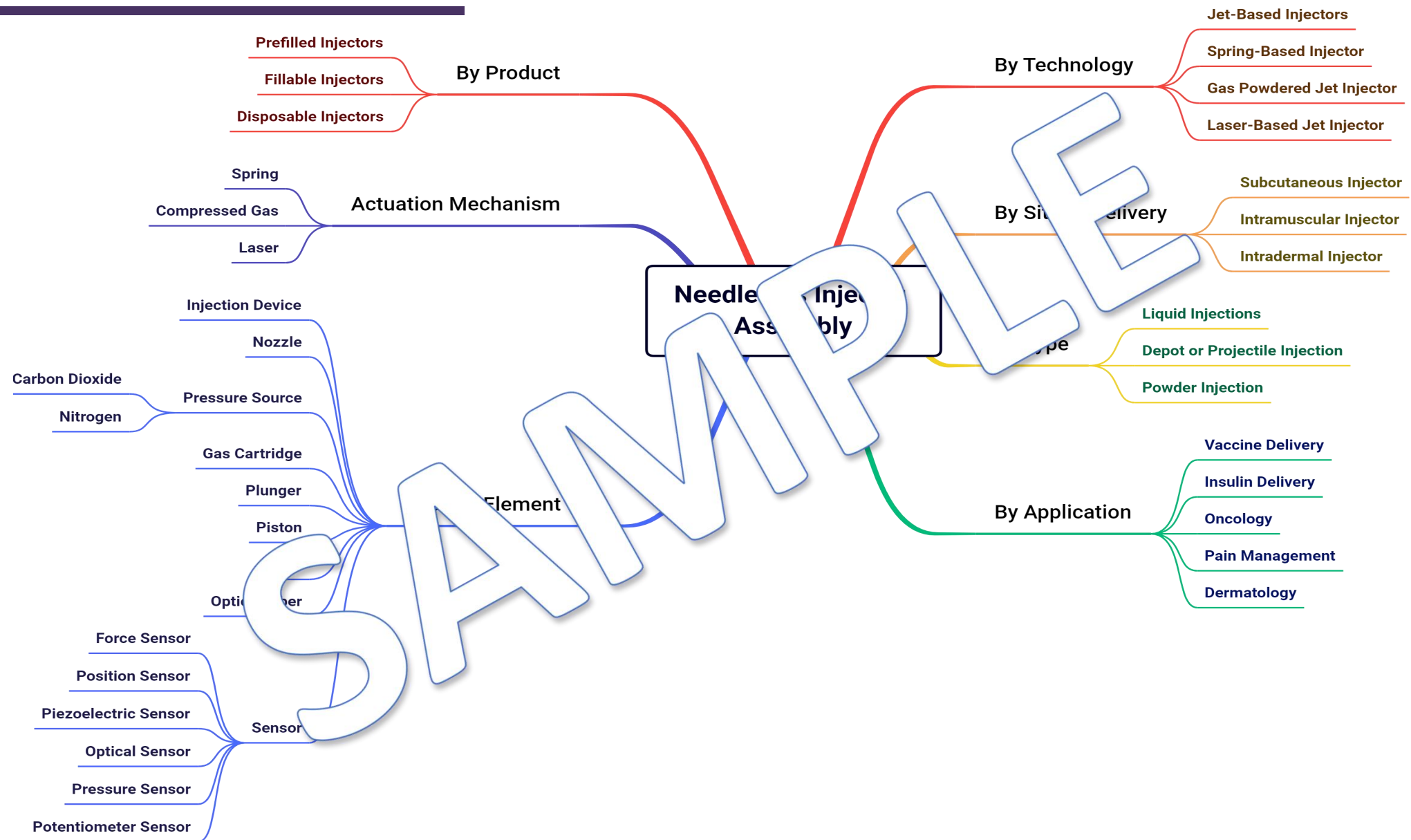


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The Present Study Provide Current Trend and active competitors insights related to Needle-Free Injector Devices.

The study will help to understand the whitespace and solutions relted to some specific problems in technology

Taxonomy

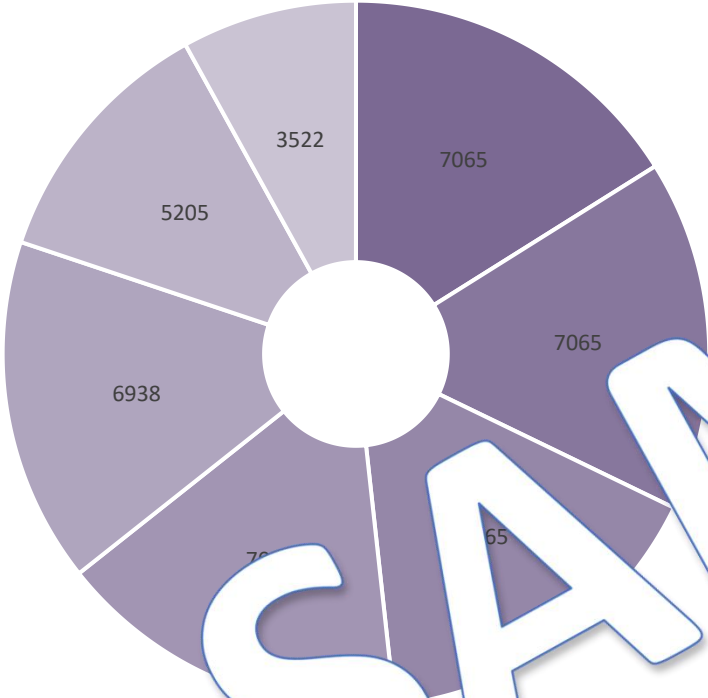


Technology Intelligence

SAMPLE

Technology Segmentation

Technology Segmentation



■ Syringe Technology ■ Actuation Mechanism ■ Injector Elements ■ By Dosage Technology ■ Injectors Type ■ Application ■ Delivery Mechanism

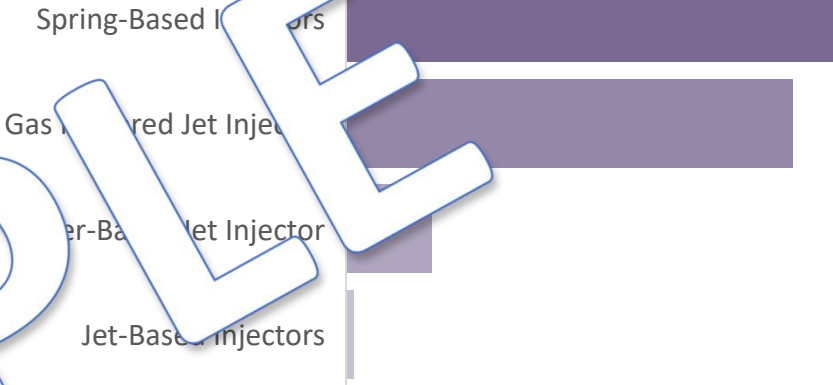
Technology Segmentation

Technology Segmentation

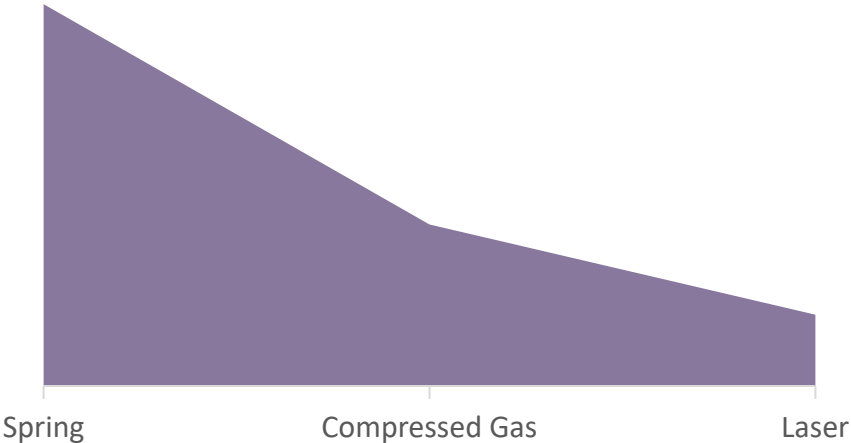
- Syringe Technology
- Actuation Mechanism
- Injector Elements
- By Dosage Technology
- Injectors Type
- Application
- Delivery Mechanism



Syringe Technology



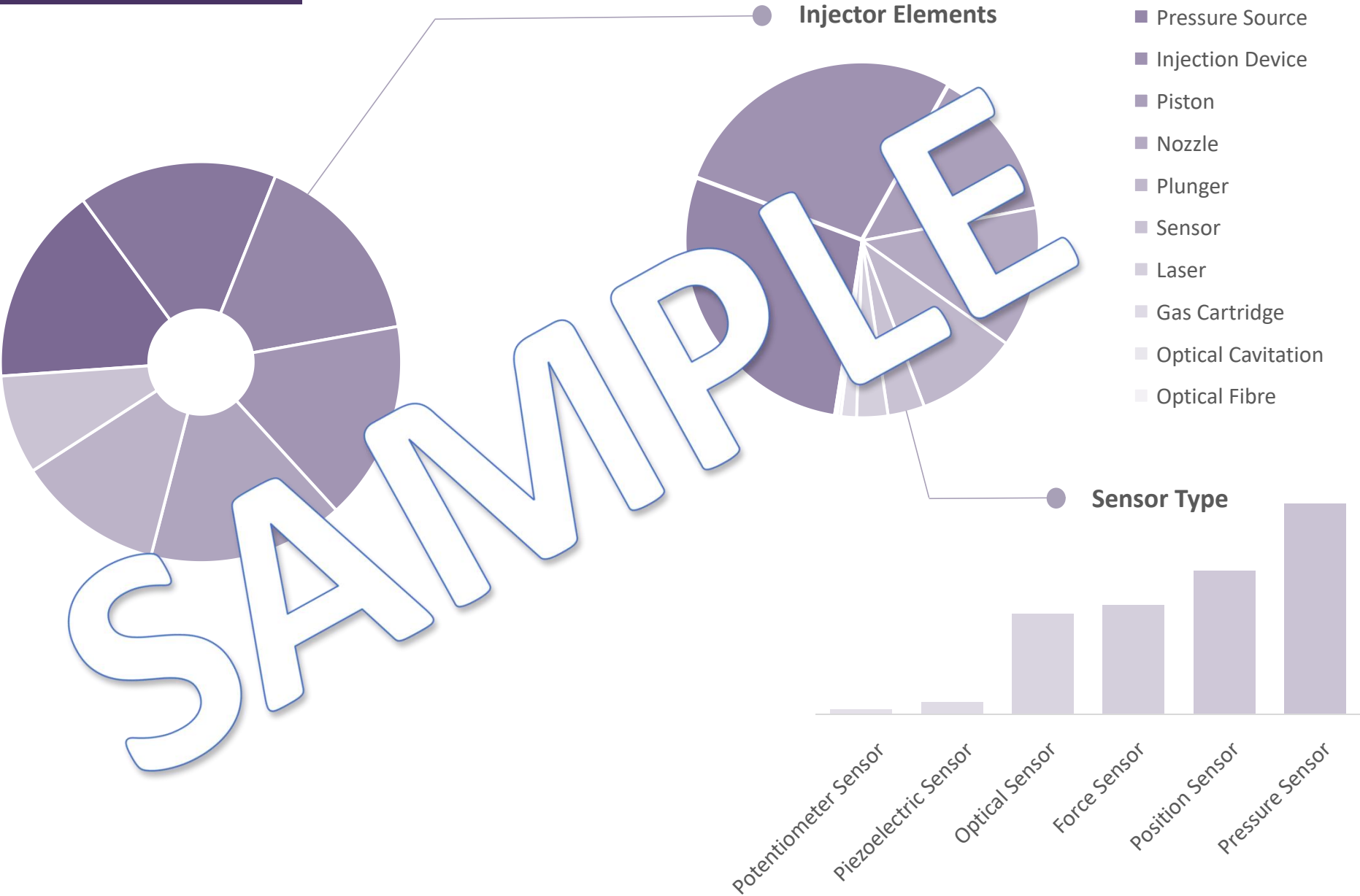
Actuation Mechanism



Technology Segmentation

Technology Segmentation

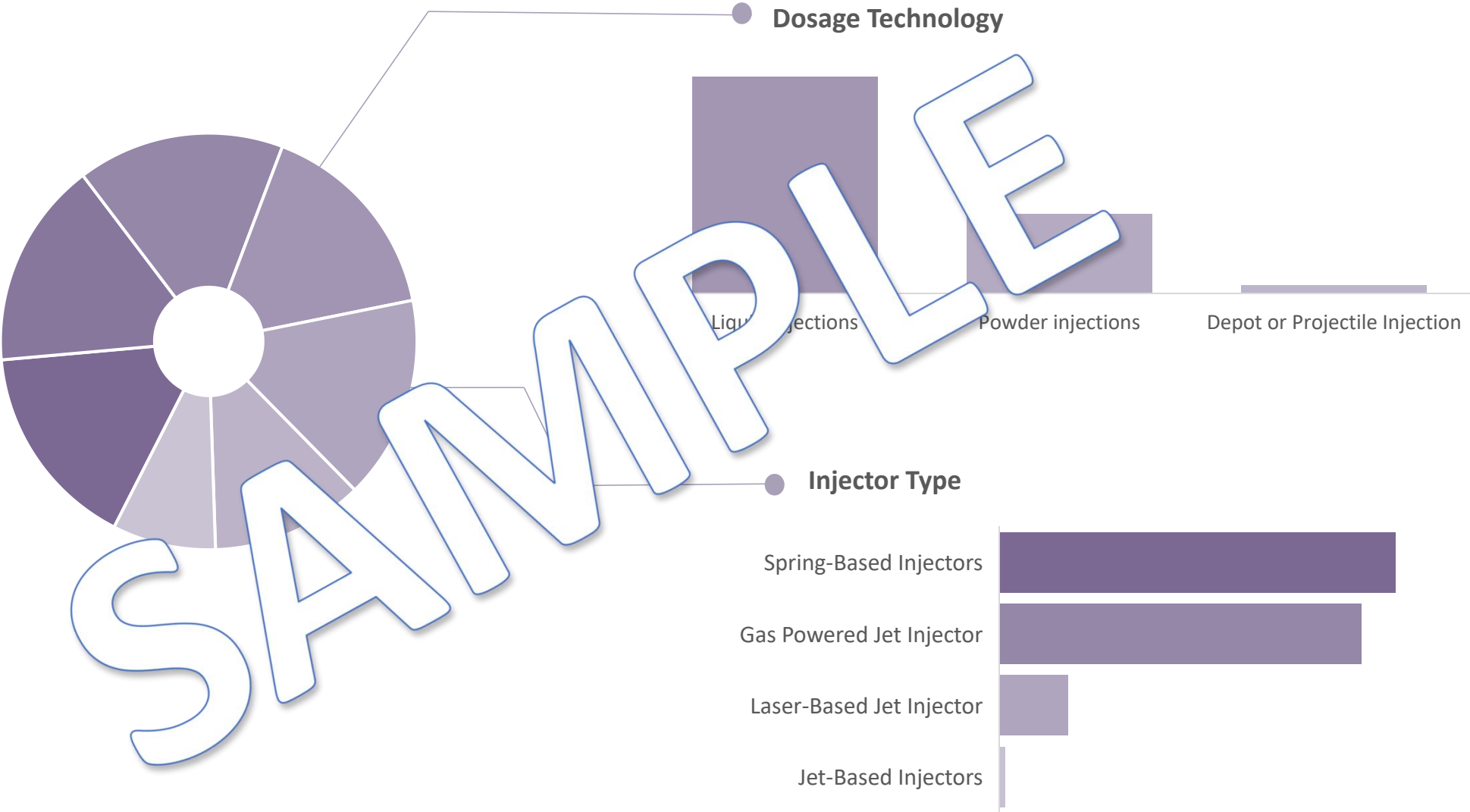
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Technology Segmentation

Technology Segmentation

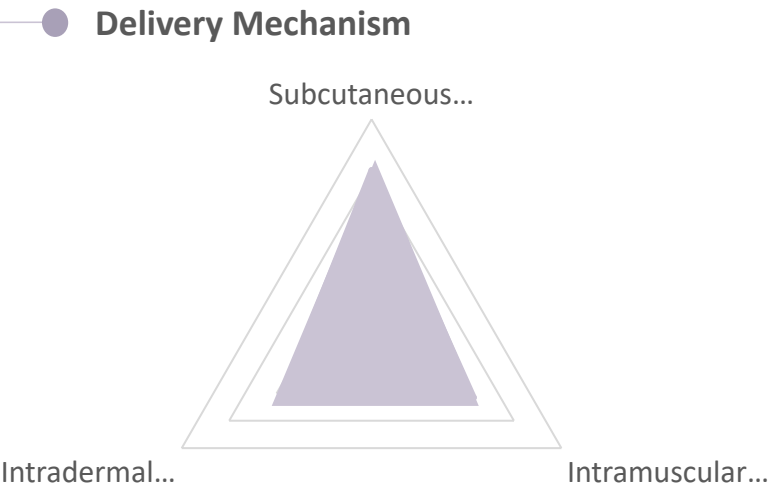
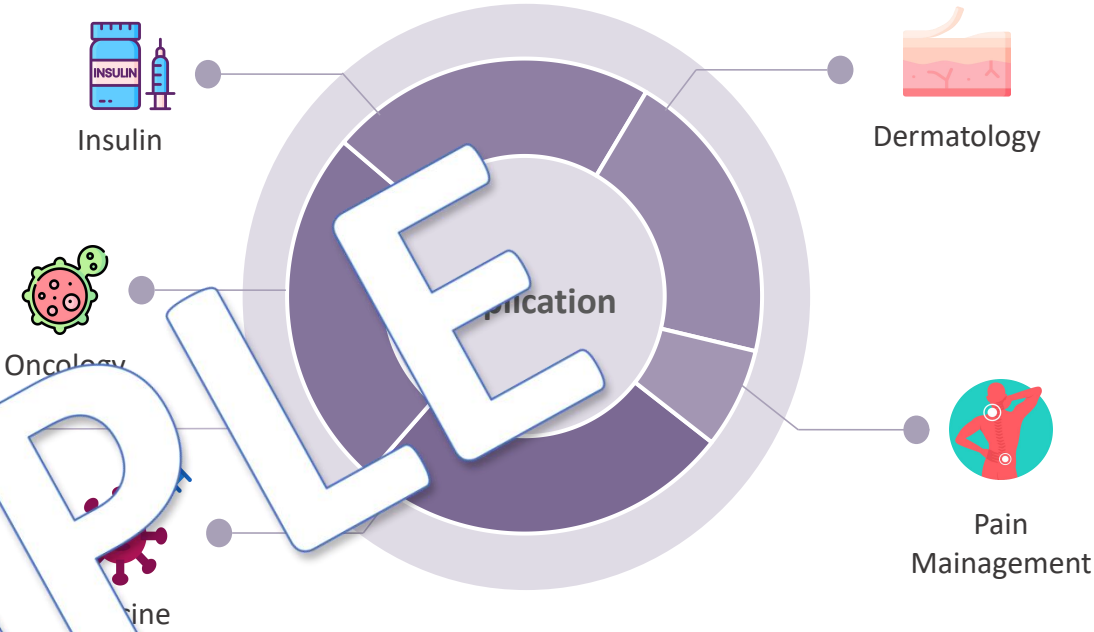
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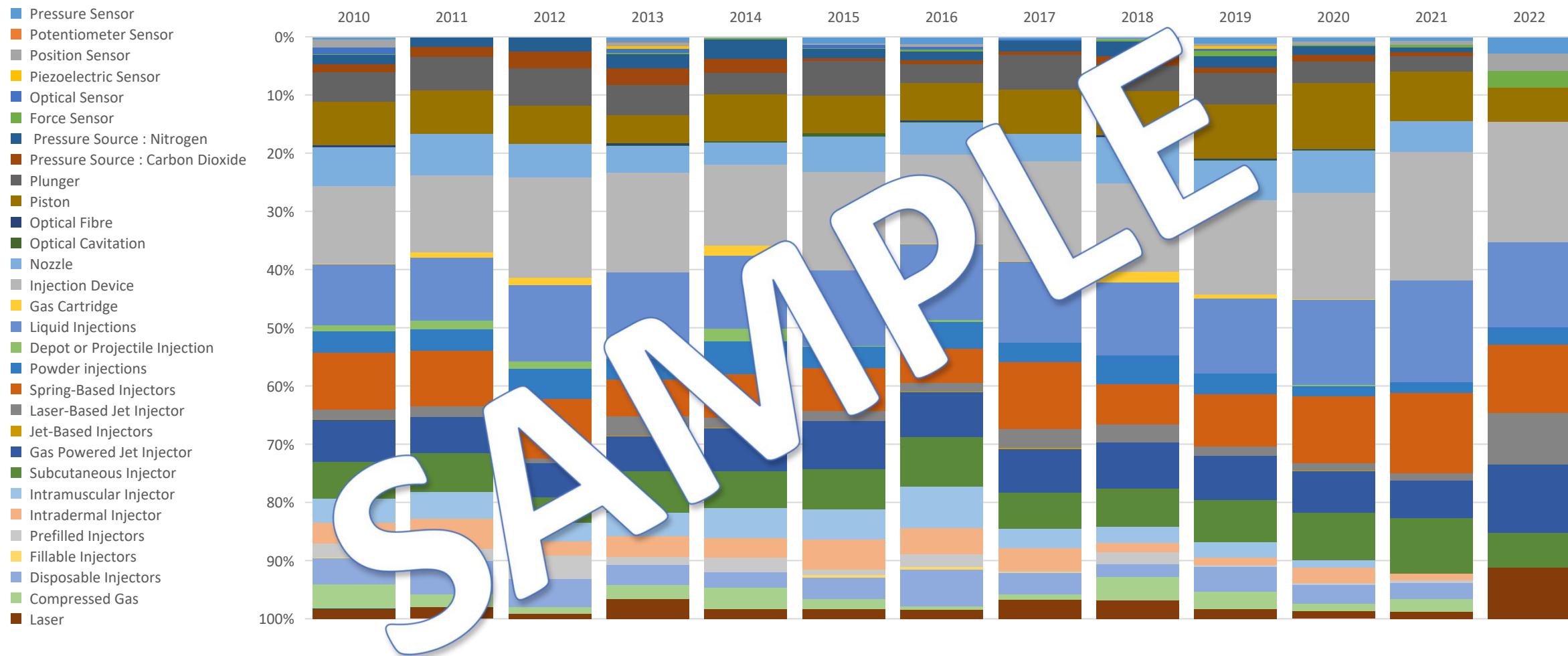


Technology Segmentation

Technology Segmentation

- Syringe Technology
- Actuation Mechanism
- Injector Elements
- By Dosage Technology
- Injectors Type
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- Delivery Mechanism





Syringe Technology	Actuation Technology		
	Spring	Laser	Compressed Gas
Jet-Based Injectors	57	3	23
Laser-Based Jet Injector	331		193
Spring-Based Injectors	3752	398	1313
Gas Powered Jet Injector	38	474	1588

Needleless Injector Technology Matrix By Application and Site of Delivery		By Technology			
		Gas Powered Jet Injector	Jet-Based Injectors	Laser Based Jet Injector	Spring-Based Injectors
By Application	Dermatology	718	125	208	706
	Insulin Delivery	767	30	208	803
	Oncology	18	312	741	
	Pain Management	4	119	247	
By Site of Delivery	Vaccination	880	10	213	880
	Intradermal	1245	10	265	1209
	Intramuscular Injector	1420	27	361	1413
	Subcutaneous Injector	1918	38	384	2019

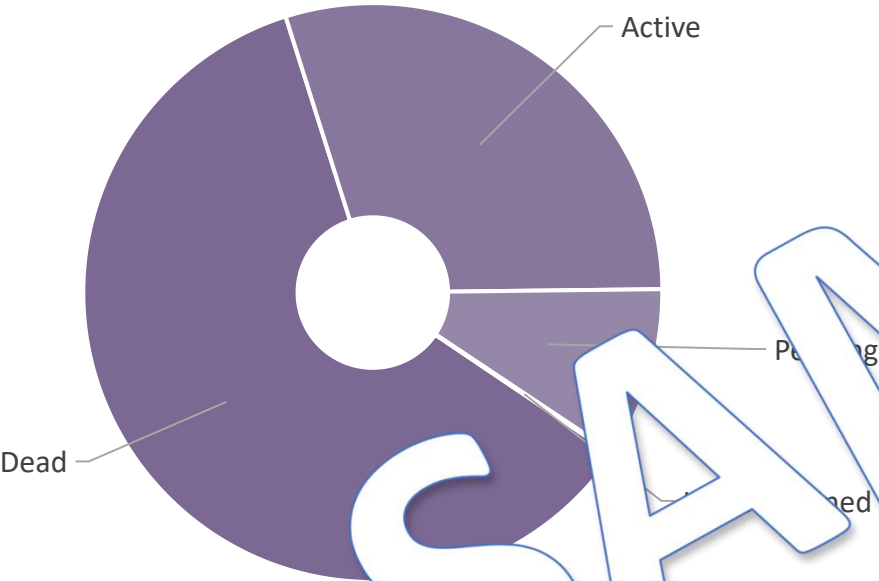
Needleless Injector Technology Matrix By Device Elements		By Technology (Millions of Units)			
		Gas Powered Jet Injector	Jet-Based Injectors	Laser-Based Jet Injectors	Spring-Based Injectors
Device Elements	Gas Cartridge	353	13	1	3132
	Injection Device	3577	64	657	3915
	Laser	474	0	702	398
	Nozzle	2271	0	319	2355
	Optical Cavitation	1	0	19	40
	Optical Fibre	0	0	56	47
	Piston	23	25	272	2813
	Plunger	1613	44	201	1753
Sensor Technology	Flow Sensor	72	1	6	88
	Optical Sensor	36	1	25	75
	Piezoelectric Sensor	12	0	5	7
	Position Sensor	103	3	13	122
	Potentiometer	2	0	2	4
	Pressure Sensor	162	1	19	136

IP Intelligence

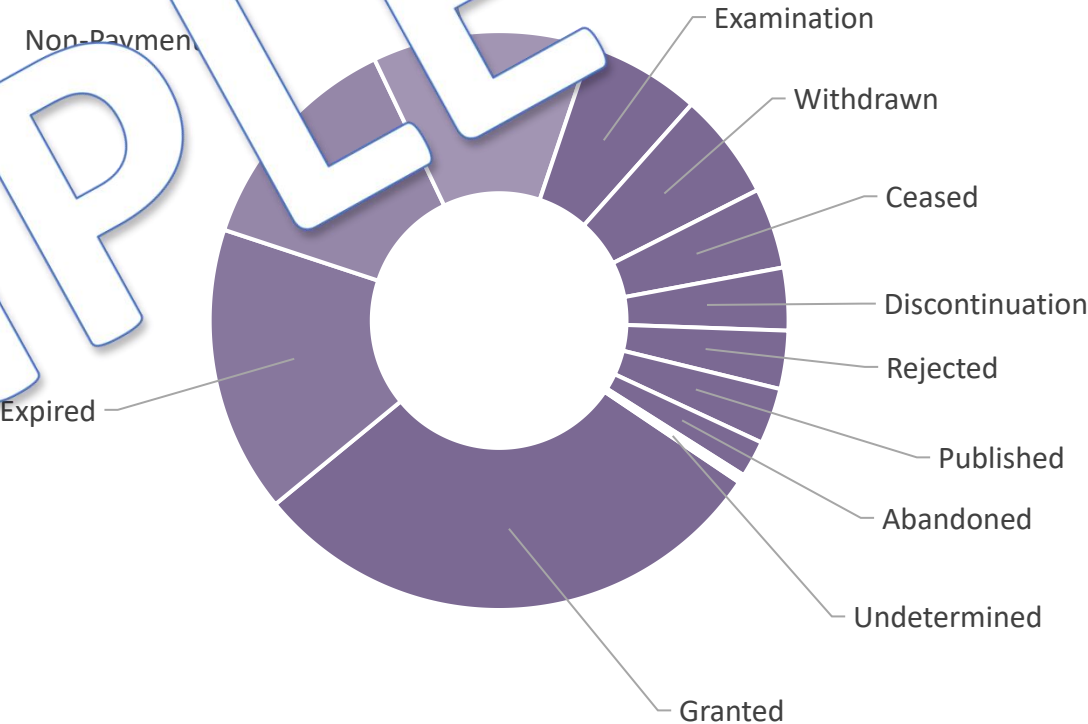
SAMPLE

Legal Status Analysis

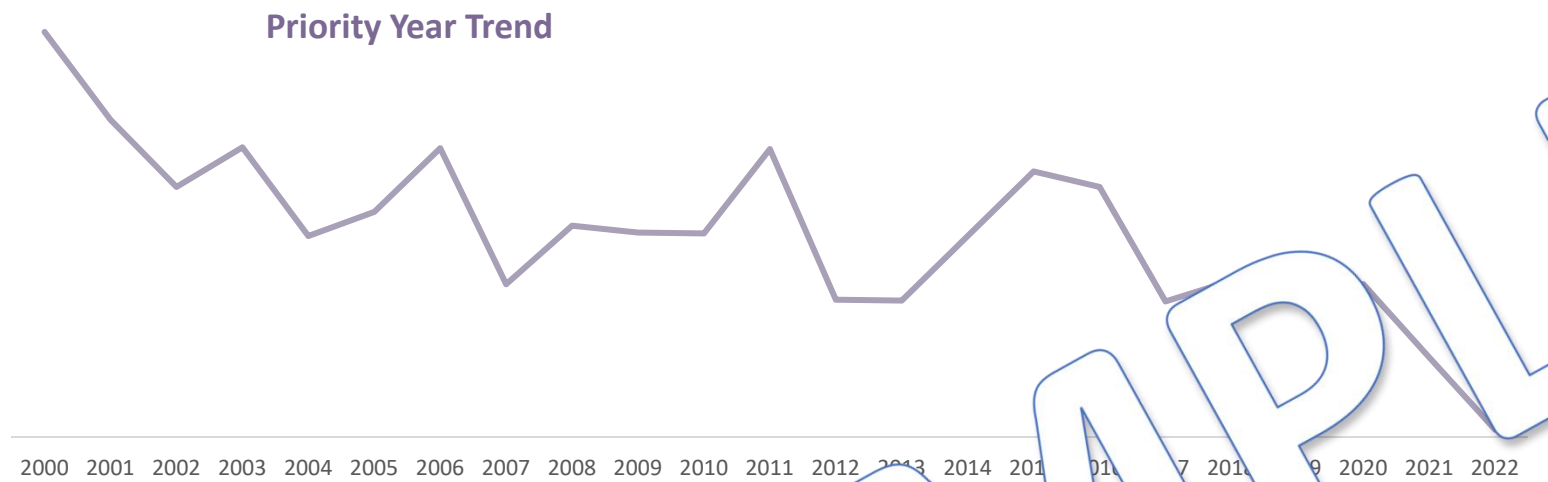
Technology Segmentation



Technology Segmentation



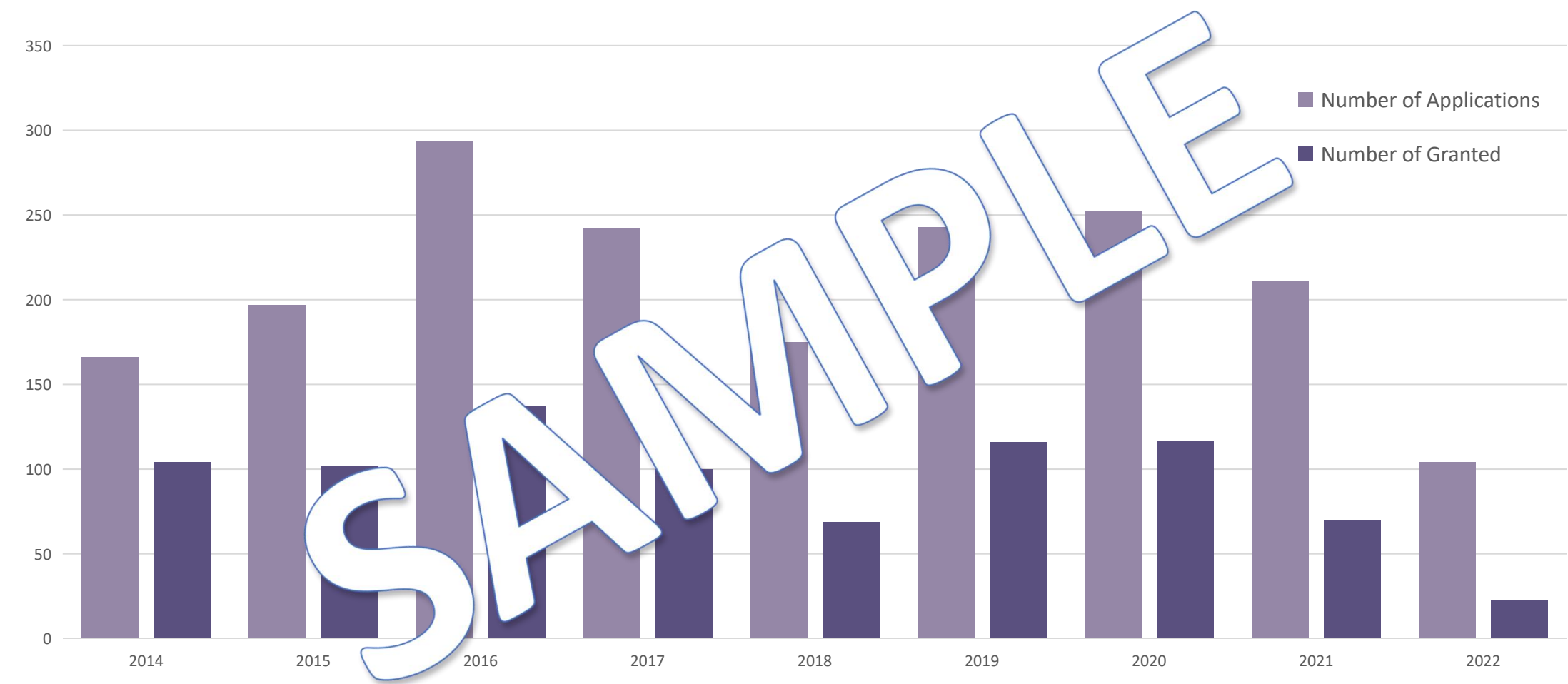
Priority And Application Trend Analysis



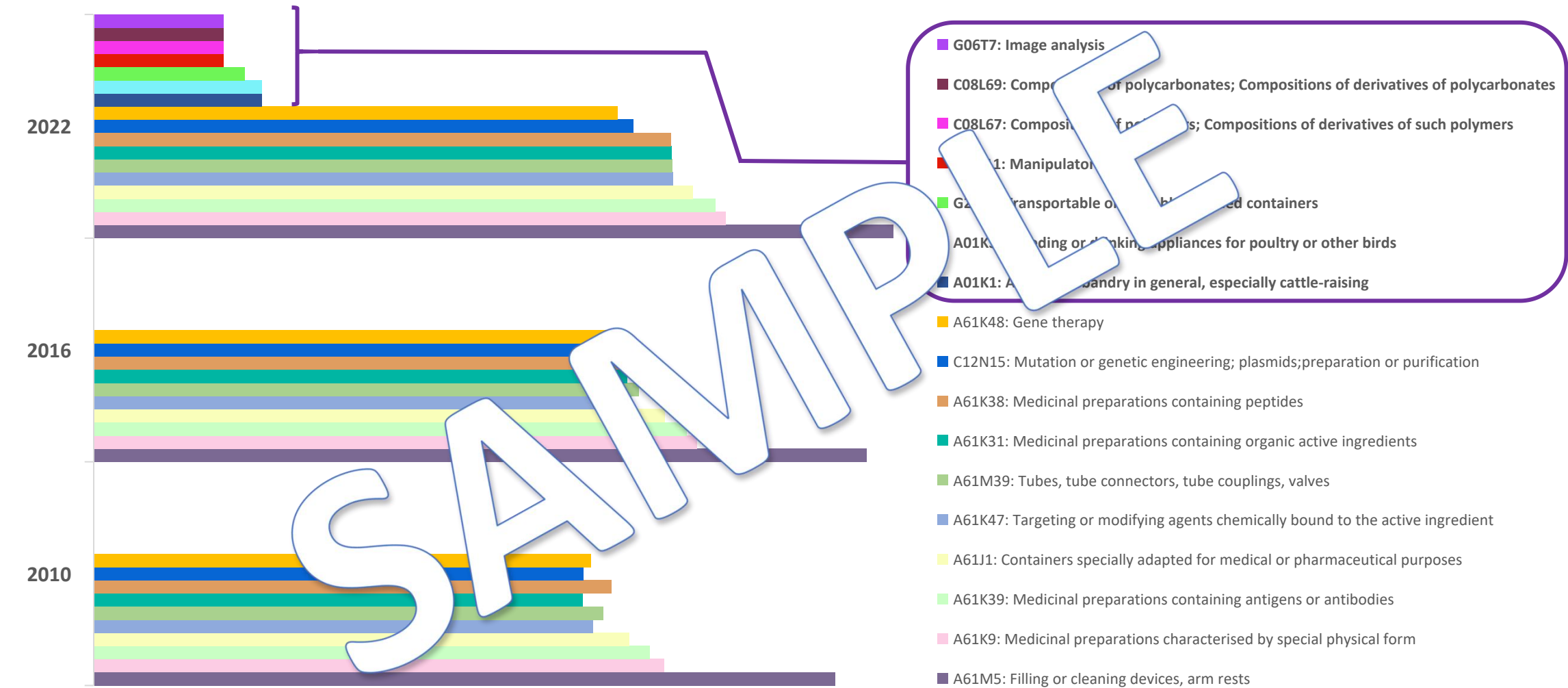
Priority Country Analysis



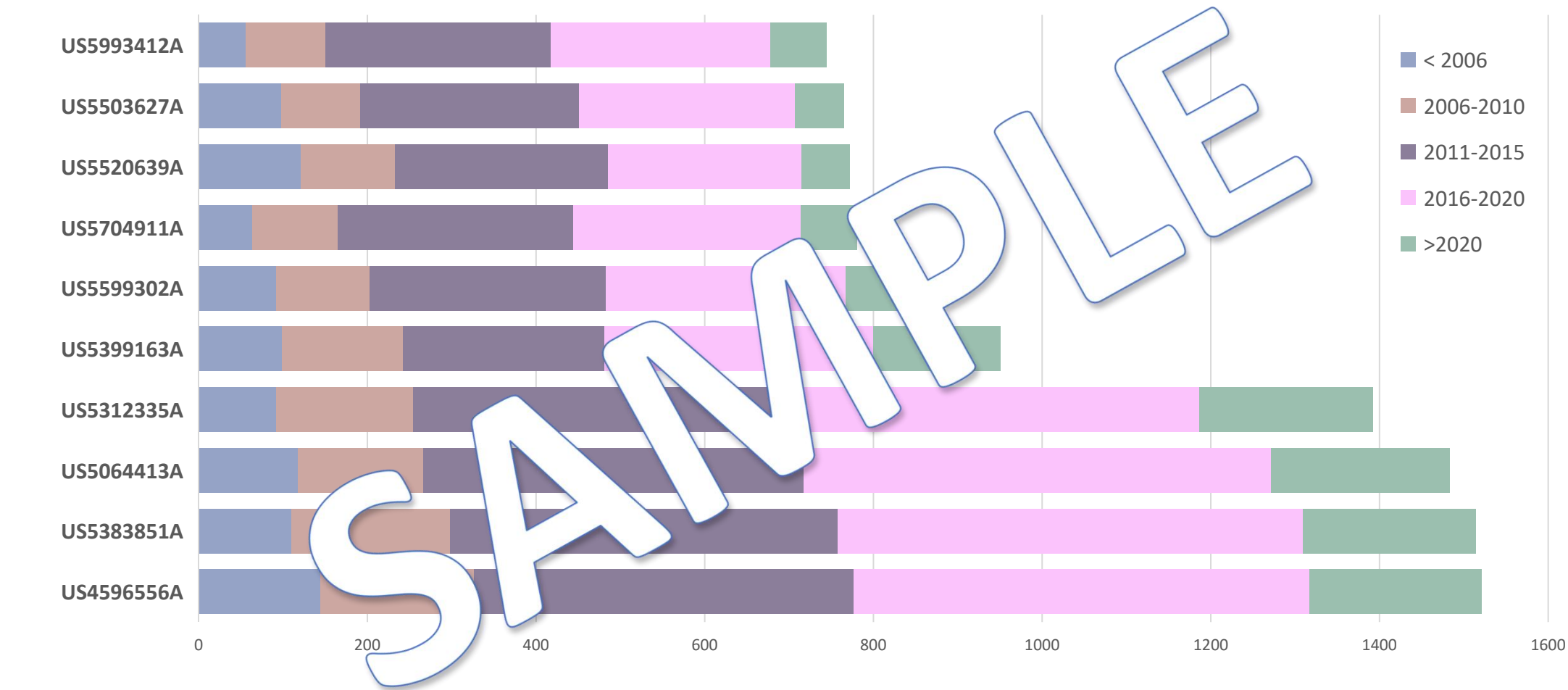
Grant Success Rate



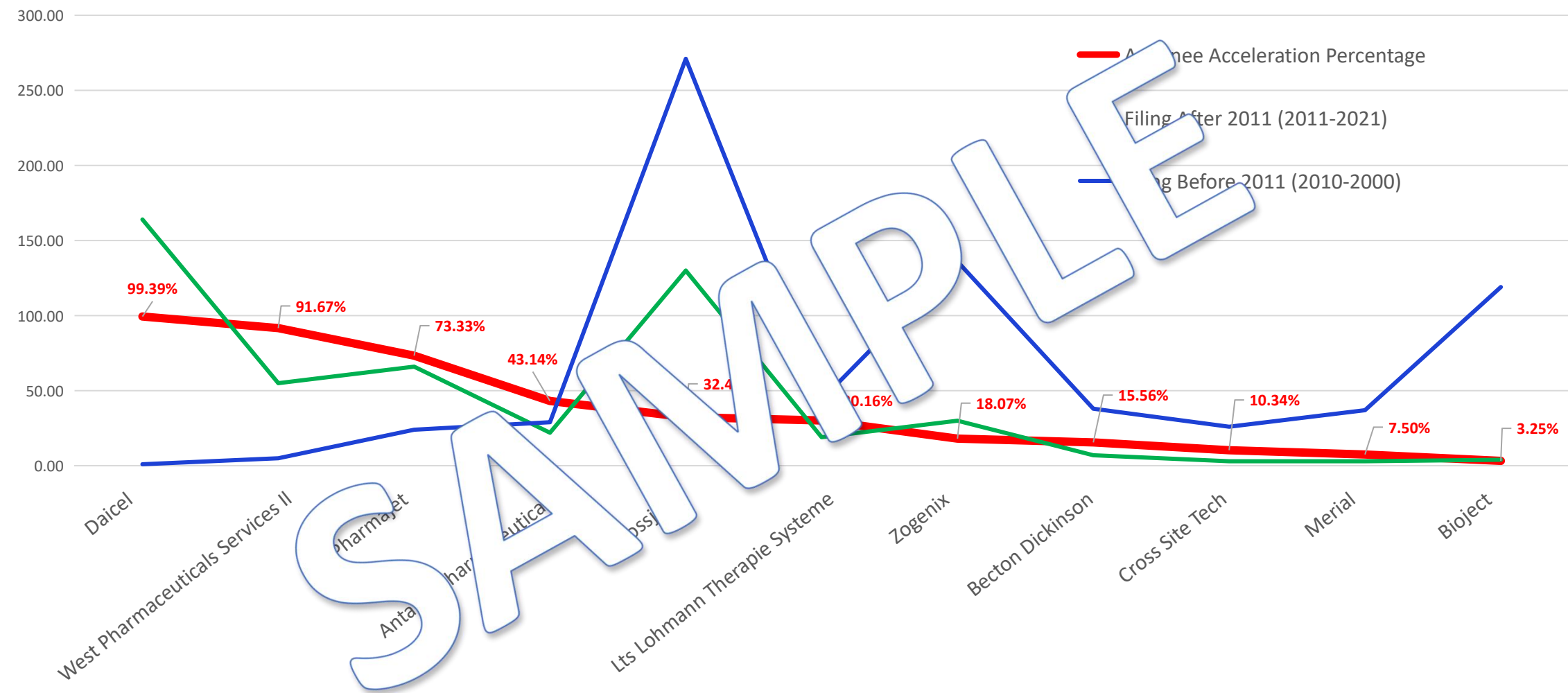
Classification Trend Analysis



Core Publication



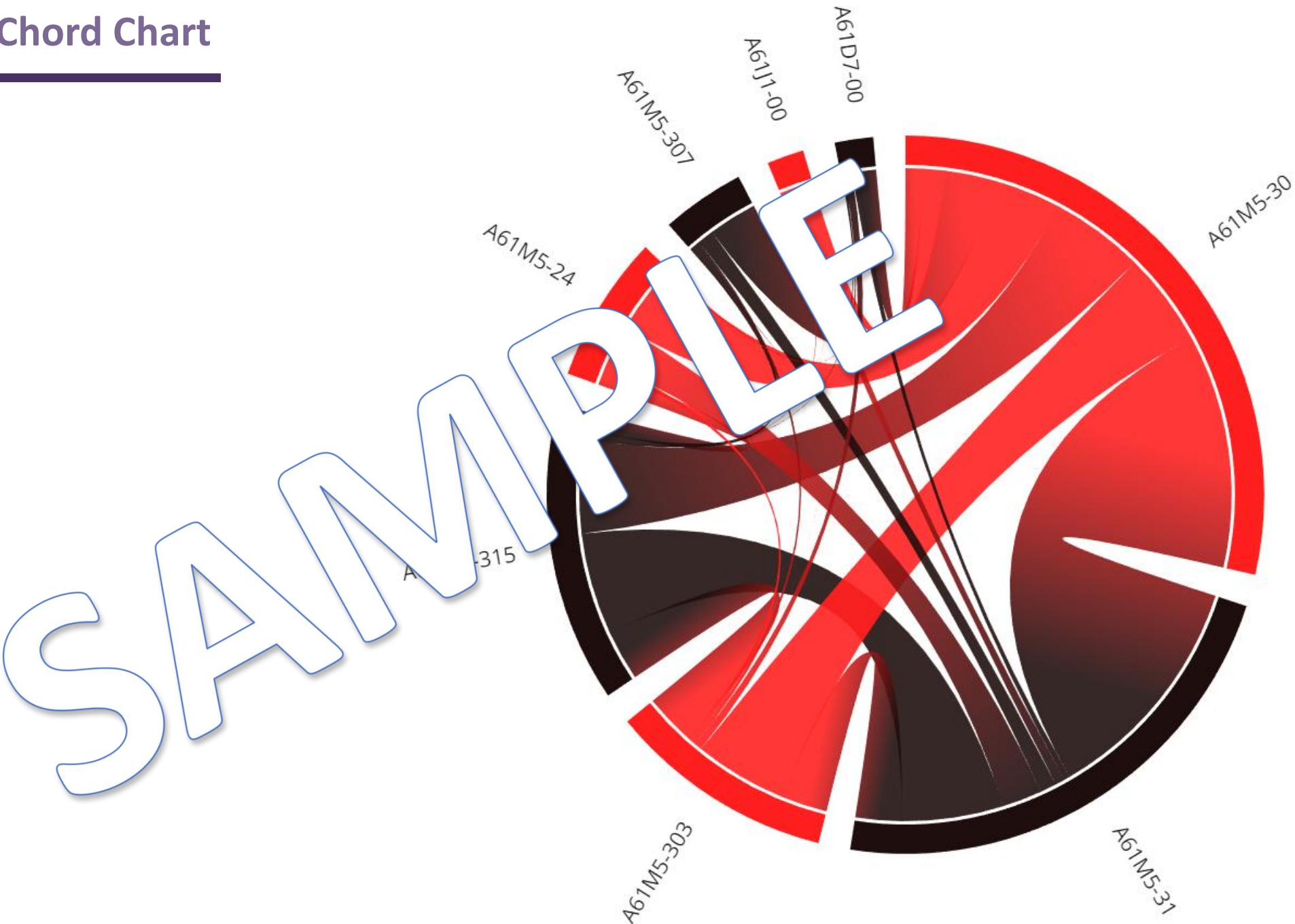
Assignee Acceleration



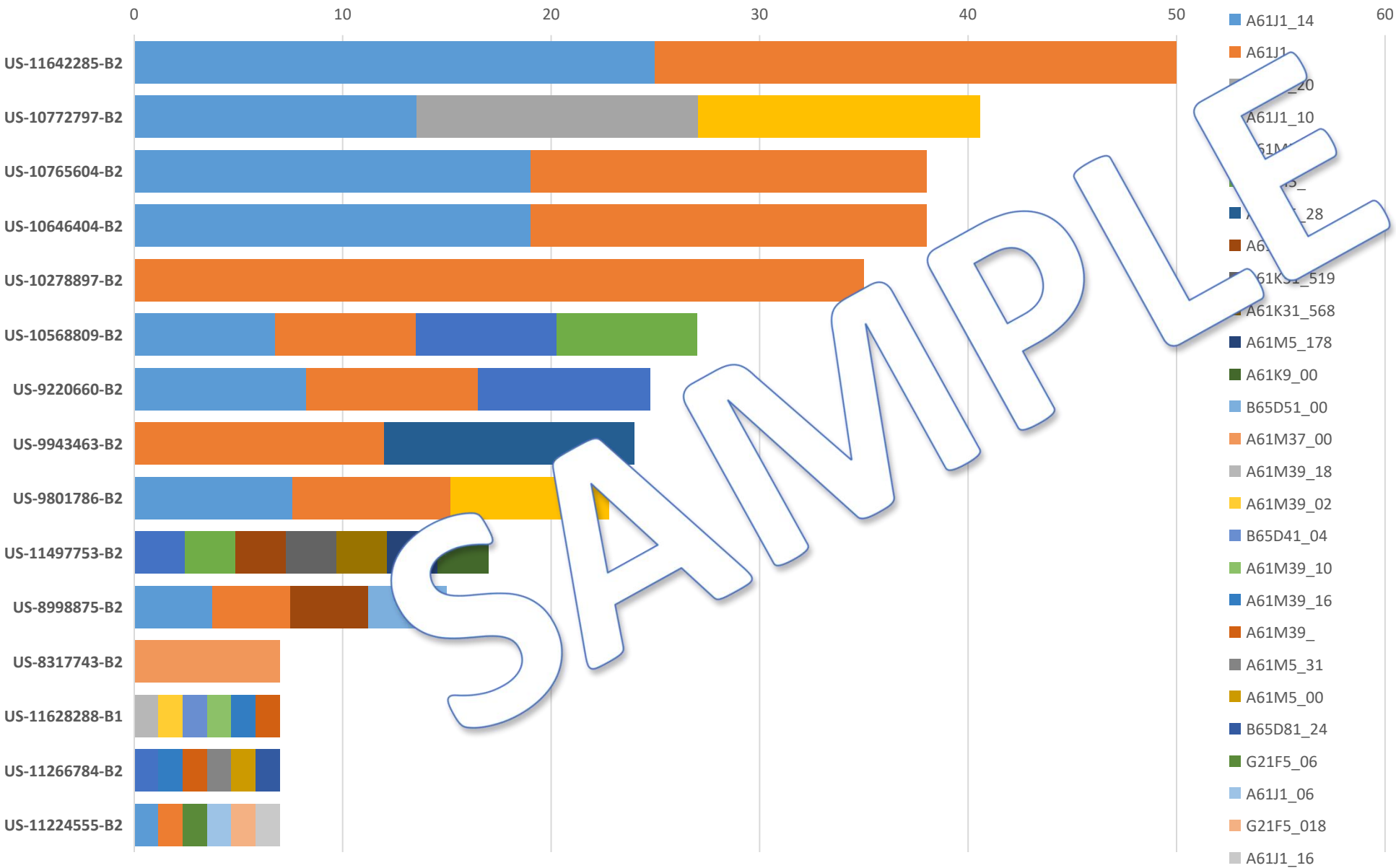
Litigation Analysis

Publication Number	Title English	Current Assignee Standardized	Earliest Priority Date	Defendant	Plaintiff
US-4966581-A	Non Reusable Disposable Capsule Containing An Individual Vaccine Dose To Be Hypodermically Injected With A Pressure Needleless Injection Apparatus	Bioject Medical Tech	1988-04-22	Merial Llc Merial Inc	Thomas A Simonian
DE-10034270-A1	Storage Container With A Metering Device For The Metered Delivery Of An Injectable Product To An Injection Device	Tecpharma Licensing Ag Bv Ch	2007-07-14	Deutsches Patent- Und Markenamt	-
BR-0012509-B1	Needleless Syringe With Injector Fitted Elements	Indres & E. Sif. Nale	1999-07-16	Inss-Instituto Nacional Do Seguro Social E Outros	Procurador Federal E Outros Fernando Campos De Arruda - Espolio E Outros
US-20100047123-A1	Sterilization Caps And Systems And Associated Methods	University Of Utah Research Foundation, Iviva Medical Systems	2007-01-16	Ivera Medical Corporation	Excelsior Medical Corporation
EP-0518561-A1	Ampule For Hypodermic Injection	Bioject	1991-06-13	Unknow	Unknow

IPC Technology Chord Chart

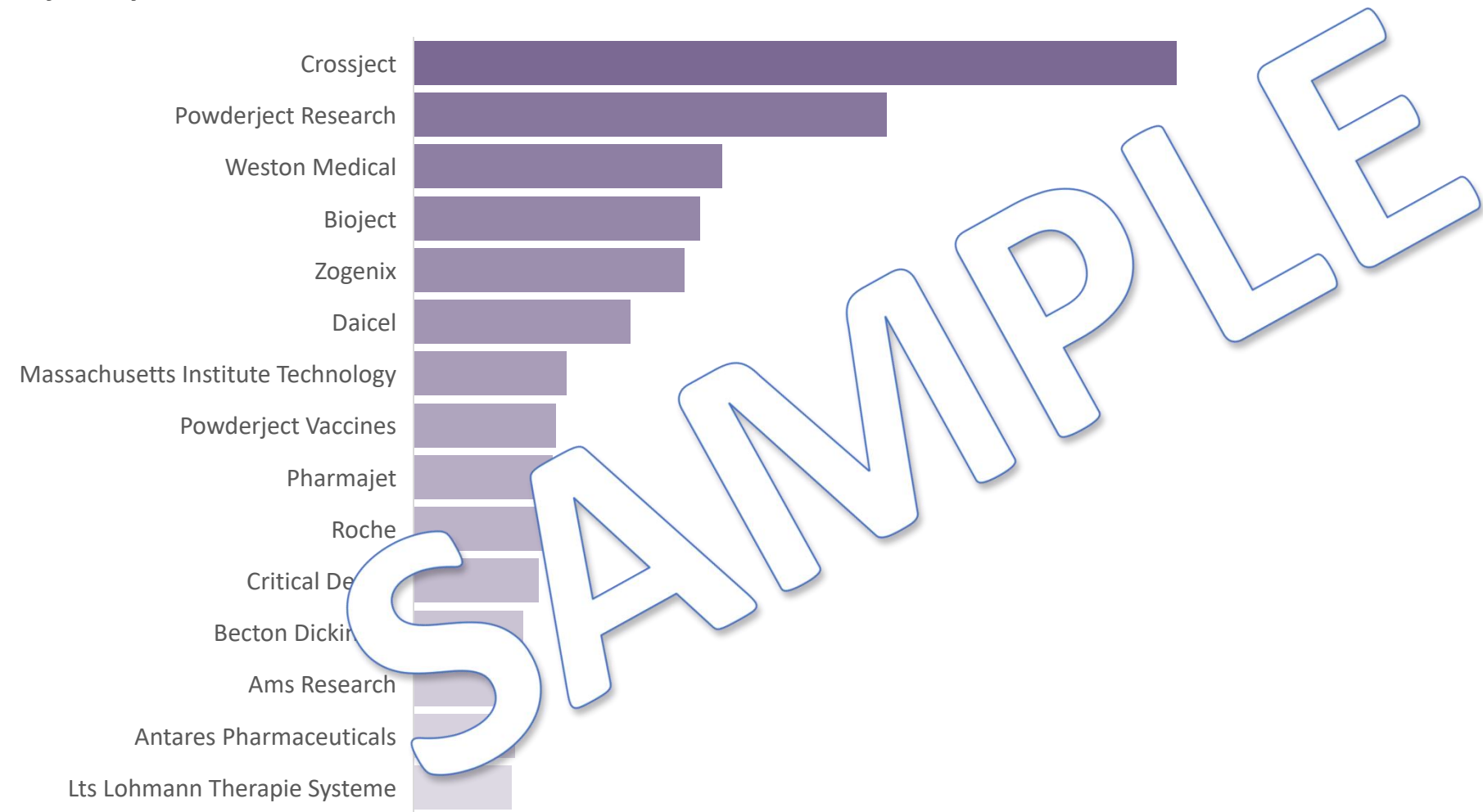


Citation Velocity



Major Players In The Domain

Major Players in IP



Competitive Intelligence

SAMPLE

Major Players In The Domain

Company name	Technology / Product Name	Description
Caretek Medical	Implaject	Simple, hand-held needle free injection device can be configured to be reusable with disposable cartridges.
 POWDERMED	PowderJect	It painlessly delivers DNA vaccine to the skin in a controlled manner.
 antares [®] pharma	Zoma-jet 2 Vision	Customized version of Microjector, now licensed to Ferring for administration of their human growth hormone for treatment of growth deficiency in Europe.
	Valeo (Microjector)	Next generation portable, spring-powered device. Designed for use with drugs in cartridge containers, rather than ampoules.
 HNS INTERNATIONAL	30	Spring-powered handheld device with disposable ampoules that delivers 0.05-0.3 ml. Focused on insulin delivery.
 Weston Medical Centre	Intraject	Applicable to drugs including proteins, peptides, monoclonal antibodies, small molecules and vaccines

Major Players In The Domain

Company name	Technology / Product Name	Description
 antares [®] pharma	Medi-Jector vision	It uses pressure to create a micro thin stream of insulin that penetrates the skin.
	Penjet	A pre filled and disposable needle free injector
Evans enterprise	Med-E- jet	Needle free injection
	Crossject	Pre-filled, single use disposable NFI. Uses chemical reaction to generate propellant at the point of administration

Competitive Analysis : Priority Trends

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Competitive Analysis : Needleless Injector Technology Segmentation

Players And Technology Matrix	Actuation Mechanism	Application	Site of Delivery	Product	Technology	Type	Injector Elements
Crossject	494	298	389	494		468	494
Powderject Research Limited	404	344	204	403	40	376	404
Zogenix Inc	162	152	103	162	162	162	162
Bioject Inc	171	115	116	171	171	167	171
Weston Medical Limited	166	89	39	166		166	166
Daicel Corp	136	116	45		136	129	136
Boehringer Ingelheim	121	88	63		121	121	121
Massachusetts Inst Technology	100	39		100	100	100	100
Roche	82	44		82	82	82	82
Pharmajet Inc	54	54	18	84	84	78	84
Lts Lohmann Therapie Systeme Ag			28	86	86	86	86
Critical Device Corp	82			82	82	82	82
Antares Pharma Inc	63	49	26	63	63	63	63
Merial Inc	50	52	49	50	50	51	50
Becton Dickinson	56	48	5	56	56	56	56

Competitive Analysis : Actuation Mechanism

Players and Actuation Mechanism Matrix	Actuation Mechanism		
	Compressed Gas	Laser	Spring
Crossject	161		291
Powderject Research Limited	204	72	86
Bioject Inc	42		115
Weston Medical Limited	50		64
Zogenix Inc	135		147
Daicel Corp	53		103
Boehringer Ingelheim	18		73
Massachusetts Inst Technology		4	76
Lts Lohmann Therapie Systeme Ag			76
Pharmajet Inc	6		70
Critical Device Corp			51
Roche			51
Antares Pharma Inc	13		57
Becton Dickinson	6	9	14
Merial Inc		1	20

Competitive Analysis : Application

Players and Application Matrix	Application				
	Insulin Delivery	Oncology	Pain Management	Vaccine Delivery	Dermatology
Powderject Research Limited	124	215	14	169	
Crossject			14		
Zogenix Inc	57	55			
Daicel Corp		78		60	
Bioject Inc	2	4		46	
Massachusetts Inst Technology	12	1	8	25	
Weston Medical Limited	8		2	13	2
Boehringer Ingelheim		5		34	1
Roche	5		1	7	
Pharmajet Inc			1	19	
Critical Device Corp					18
Lts Lohmann Therapie Systeme Ag	3	1			
Merial Inc		11		52	1
Ams Research		2			
Antares Pharma Inc	24	14	2	8	

Competitive Analysis : Product Type

Players and Product Type Matrix	Product Type		
	Disposable Injectors	Fillable Injectors	Prefilled Injectors
Crossject	323		115
Zogenix Inc	117		89
Powderject Research Limited	128		33
Bioject Inc	76		33
Lts Lohmann Therapie Systeme Ag	75	10	10
Weston Medical Limited	59		27
Roche	57		28
Massachusetts Inst Technology	58		7
Shire Human Genetics Therapies Inc			36
Primojex	26	8	24
Pharmajet Inc	22		22
Ams Research	45		
Antares Pharma Inc	10	14	19
Genentech Inc	22		16
Idee Internat R&D Inc	35		3

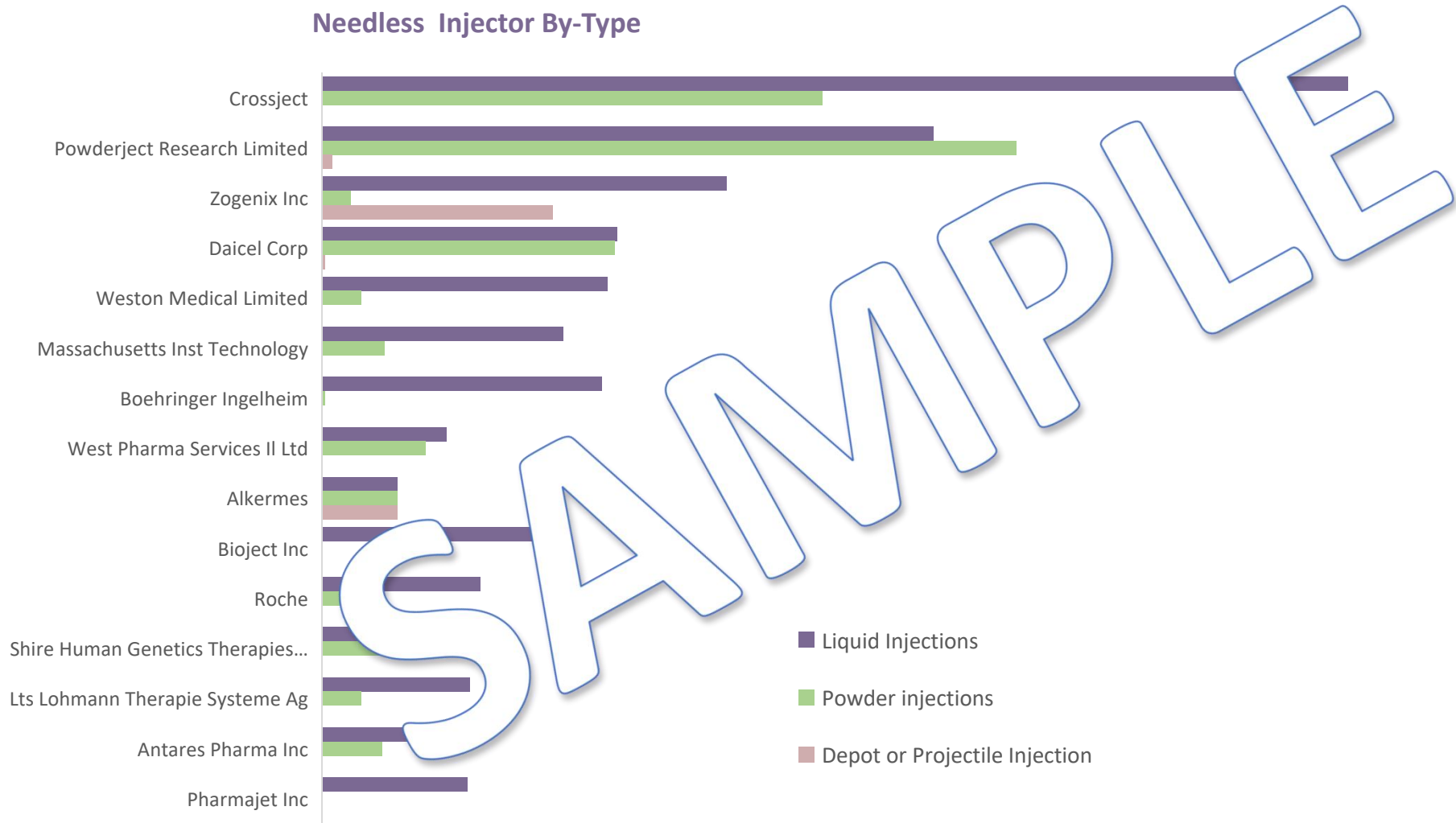
Competitive Analysis : Site of Delivery

Players and Product Site of Delivery	Site of Delivery		
	Intradermal Injector	Intramuscular Injector	Subcutaneous Injector
Crossject	337	385	356
Powderject Research Limited	152	122	102
Bioject Inc	92		112
Zogenix Inc	75	9	102
Boehringer Ingelheim	26		56
Merial Inc	43	48	49
Pharmajet Inc	42		17
Daicel Corp	28	1	29
Massachusetts Inst Technology		9	17
Roche		8	44
Shire Human Genetics Therapies Inc		38	40
Weston Medical Limited		7	35
Acushot Inc			36
Shimadzu Corp		1	32
Primojex	14	20	30

Competitive Analysis : Technology

Players and Technology Type	Technology			
	Gas Powered Jet Injector	Jet-Based Injectors	Laser-Based Jet Injector	Spring-Based Injectors
Crossject	411	1		291
Powderject Research Limited	376		72	140
Bioject Inc	101	1	11	116
Weston Medical Limited	84			65
Zogenix Inc	152		18	147
Daicel Corp	115			103
Boehringer Ingelheim	66		1	73
Massachusetts Inst Technology	74	3	4	87
Lts Lohmann Therapie Systeme Ag	41			76
Pharmajet Inc	33	2		70
Critical Device Corp				51
Roche	2			57
Antares Pharma Inc		22		57
Becton Dickinson	12		9	14
Merial Inc	1		1	20

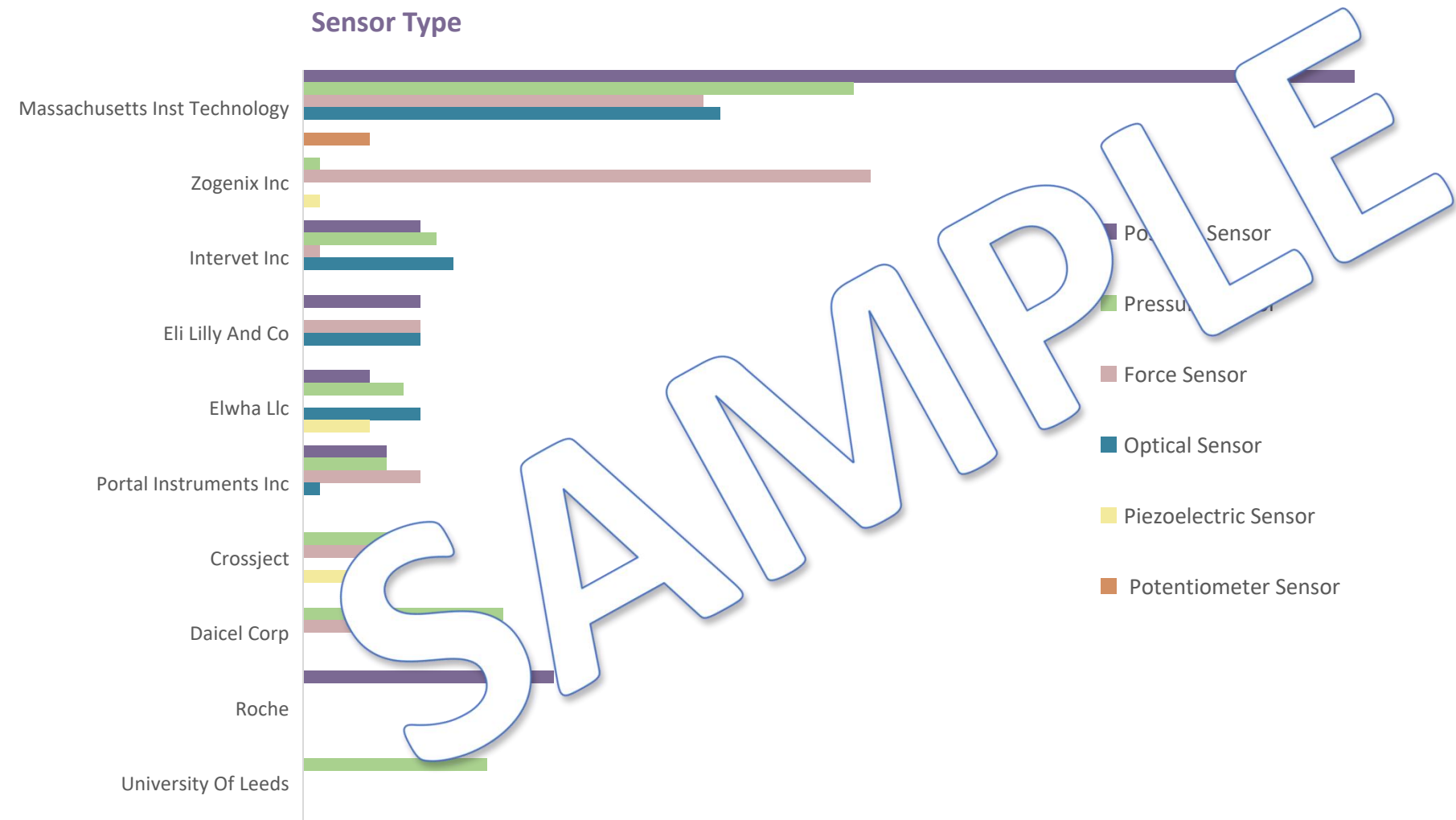
Competitive Analysis : Needleless Injector By-Type

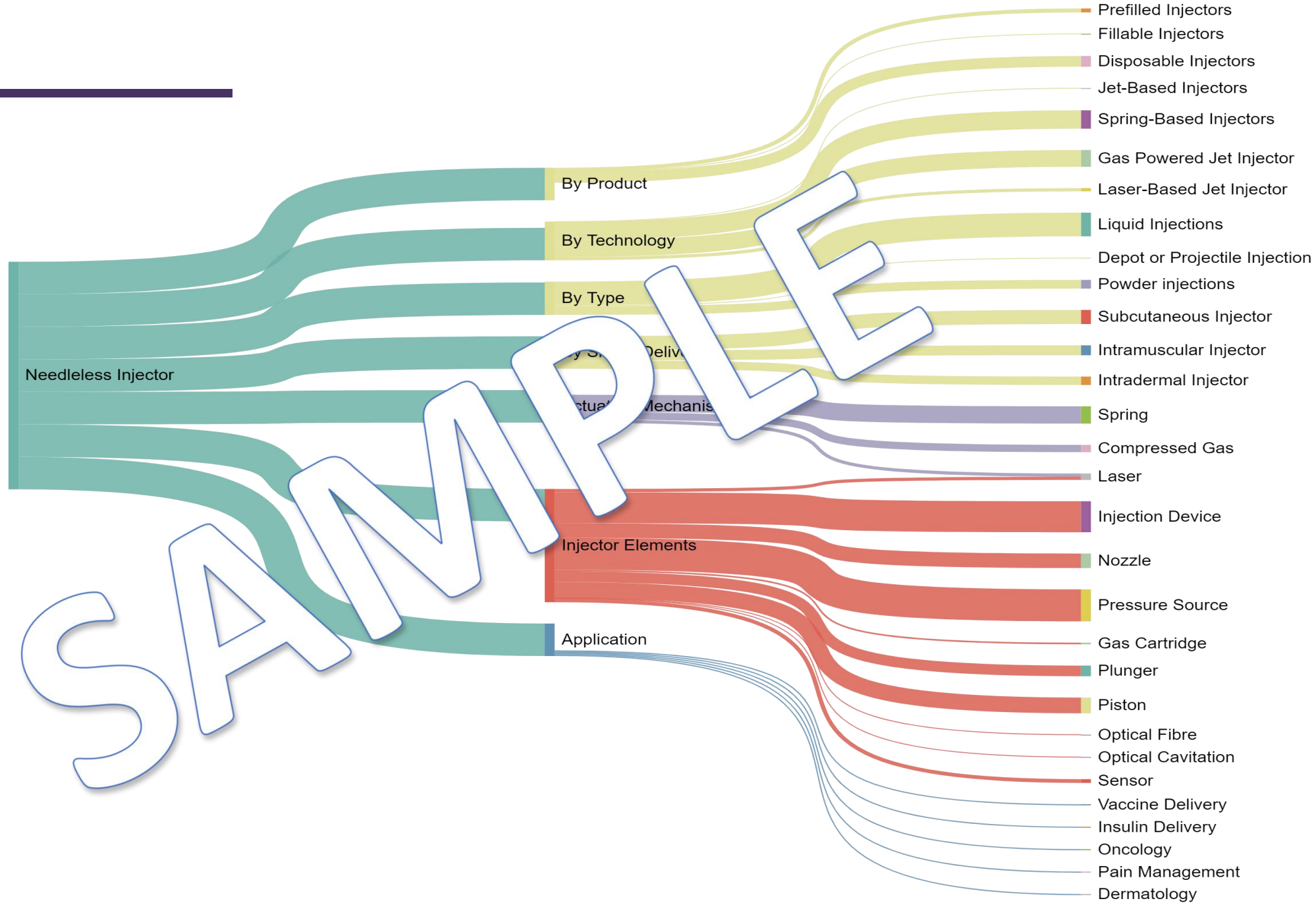


Competitive Analysis : Injector Element

Assignee/Applicant	Pressure Source		Injection Device	Piston	Nozzle	Plunger	Sensor	er	Gas Cartridge	Optical Fibre	Optical Cavitation
	Carbon Dioxide	Nitrogen									
Crossject		22	494	316	244	138	18		14		
Powderject Research Limited	52	162	402	77	346		12		23		
Zogenix Inc	96	30	158	145	121	10	7		21		
Bioject Inc	32	26	159	84	136			11	55		
Daicel Corp			135	115	14	74	17		6		
Massachusetts Inst Technology	2	1	10	91			84	4		2	14
Weston Medical Limited	2	8	150	36		41	13	2	3		
Boehringer Ingelheim	21	7	116	81	80	34		1	3		
Roche		16	1	9	49	50	27		20		1
Pharmajet Inc			34	66	68	65					
Lts Lohmann Therapie Systeme Ag			84	78	65	27			6		
Ams Research	6		50	45	37	45	9	14	4	16	
Acushot Inc	36	36	44	40	36	37					
Antares Pharma Inc		2	63	15	57	53			10		
Becton Dickinson		28	55	47	3	45		9			

Competitive Analysis : Injector Element : Sensor Type





Whitespace & Key Takeaways

Potential Identified Gaps In Needleless Injector Technology

